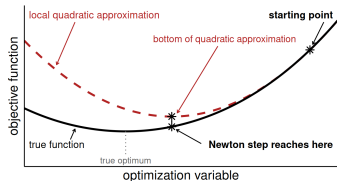


Optimization in Machine Learning

Second order methods

Newton-Raphson



Learning goals

- Newton-Raphson: Motivation and derivation
- Damped Newton-Raphson
- Affine invariance of the Newton step

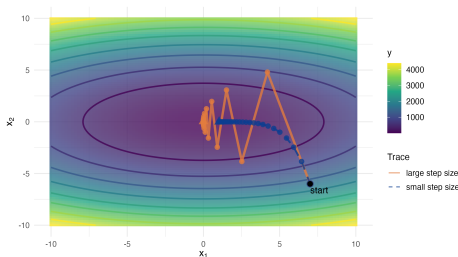
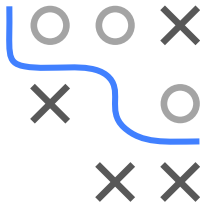
Slides based on

► Aggarwal 2020 (Ch. 5.4-5.7)

► Boyd and Vandenberghe 2004 (Ch. 9.5)

MOTIVATION: BEYOND GRADIENT INFORMATION

- **Recall:** GD only uses the gradient $\Rightarrow$ *ignores curvature*
- Hessian $\mathbf{H} = \nabla^2 f$ (curvature) encodes how fast the gradient itself changes along each direction
 - High curvature: gradient is unreliable further away $\Rightarrow$ take a *small* step
 - Low curvature: gradient stays valid further away $\Rightarrow$ take a *large* step
- GD only picks one global α for all directions $\Rightarrow$ for ill-conditioned $\mathbf{H}$, this causes zig-zagging / slow convergence:



A 3x3 grid with a blue path starting at the top-left corner and ending at the bottom-right corner. The path passes through the middle-right cell. The grid contains several 'X' marks and one 'O' mark.

-

Effect of multiplying the steepest-descent direction with the inverse Hessian

NEWTON-RAPHSON

- **Assumption:** $f \in \mathcal{C}^2$
- **Aim:** Find stationary point $\mathbf{x}^*$, i.e., $\nabla f(\mathbf{x}^*) = \mathbf{0}$
- **Idea:** Find root of first order Taylor approx of $\nabla f(\mathbf{x})$:

$$\nabla f(\mathbf{x}) \approx \nabla f(\mathbf{x}^{[t]}) + \nabla^2 f(\mathbf{x}^{[t]})(\mathbf{x} - \mathbf{x}^{[t]}) = \mathbf{0}$$

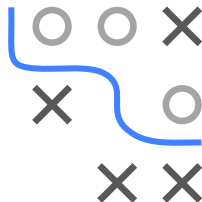
$$\nabla^2 f(\mathbf{x}^{[t]})(\mathbf{x} - \mathbf{x}^{[t]}) = -\nabla f(\mathbf{x}^{[t]})$$

$$\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} - (\nabla^2 f(\mathbf{x}^{[t]}))^{-1} \nabla f(\mathbf{x}^{[t]})$$

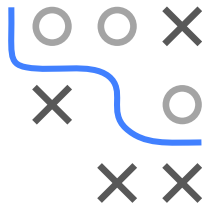
- **Update scheme:**

$$\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} + \mathbf{d}^{[t]}$$

$$\text{with } \mathbf{d}^{[t]} = -(\nabla^2 f(\mathbf{x}^{[t]}))^{-1} \nabla f(\mathbf{x}^{[t]})$$



NEWTON-RAPHSON: PRACTICE



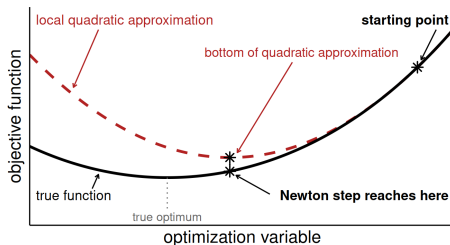
- **Note:** In practice, we get $\mathbf{d}^{[t]}$ by solving the linear system

$$\nabla^2 f(\mathbf{x}^{[t]})\mathbf{d}^{[t]} = -\nabla f(\mathbf{x}^{[t]})$$

with direct (matrix decompositions) or iterative methods

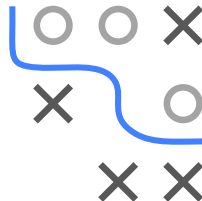
NEWTON-RAPHSON: INTUITION

- Newton update approximates the function locally with a quadratic bowl and moves to the bottom of the bowl in a single step
- The “learning rate” is incorporated implicitly
- In general, the bottom of the quadratic bowl is not the bottom of the true function $\Rightarrow$ multiple Newton updates are usually necessary



Note: univariate functions are rather “boring” as there are only two possible directions $\Rightarrow$ we’ll move on to more interesting examples soon

- **Caveat:** if the quadratic approximation is poor (e.g., far from the current point), the Newton step may worsen f (see later examples)



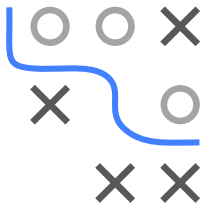
ANALYTICAL EXAMPLE WITH QUADRATIC FORM

- Consider $f(x_1, x_2) = x_1^2 + \frac{x_2^2}{2}$
- Update direction: $\mathbf{d}^{[t]} = -(\nabla^2 f(x_1^{[t]}, x_2^{[t]}))^{-1} \nabla f(x_1^{[t]}, x_2^{[t]})$
- Gradient and Hessian:

$$\nabla f(x_1, x_2) = \begin{pmatrix} 2x_1 \\ x_2 \end{pmatrix}, \quad \nabla^2 f(x_1, x_2) = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

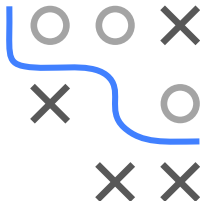
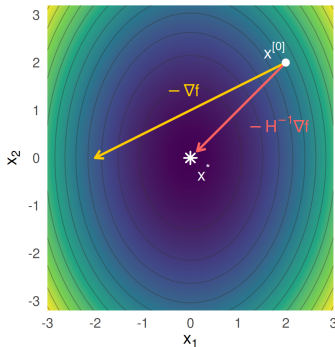
- First step:

$$\begin{aligned} \begin{pmatrix} x_1^{[1]} \\ x_2^{[1]} \end{pmatrix} &= \begin{pmatrix} x_1^{[0]} \\ x_2^{[0]} \end{pmatrix} + \mathbf{d}^{[0]} = \begin{pmatrix} x_1^{[0]} \\ x_2^{[0]} \end{pmatrix} - \begin{pmatrix} 1/2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2x_1^{[0]} \\ x_2^{[0]} \end{pmatrix} \\ &= \begin{pmatrix} x_1^{[0]} \\ x_2^{[0]} \end{pmatrix} + \begin{pmatrix} -x_1^{[0]} \\ -x_2^{[0]} \end{pmatrix} = \mathbf{0} \end{aligned}$$



ANALYTICAL EXAMPLE WITH QUADRATIC FORM

- Newton-Raphson only needs one iteration for quadratic forms
- See how the inverse Hessian rescales the gradient:
 - x_1 : higher curvature $\Rightarrow$ scaled by $1/2$ by $\mathbf{H}^{-1}$
 - x_2 : lower curvature $\Rightarrow$ scaled by 1 by $\mathbf{H}^{-1}$
- **Note:** For diagonal $\mathbf{H}$, each dimension is scaled individually and we can easily interpret the scaling factors; for non-diagonal $\mathbf{H}$, the update direction becomes a linear combination of all dimensions

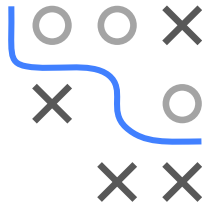
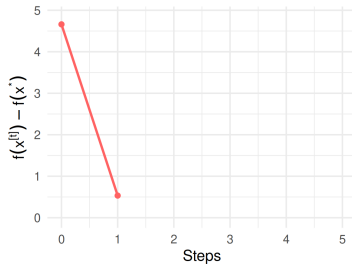
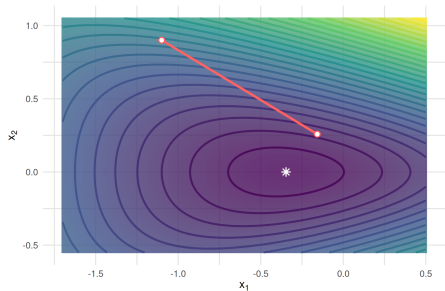


NON-QUADRATIC EXAMPLES: CONVERGENCE

Example 1: Newton-Raphson on

$$f(\mathbf{x}) = \exp(x_1 + 3x_2 - 0.1) + \exp(x_1 - 3x_2 - 0.1) + \exp(-x_1 - 0.1)$$

After 1 Newton step:

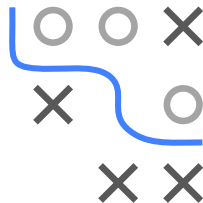
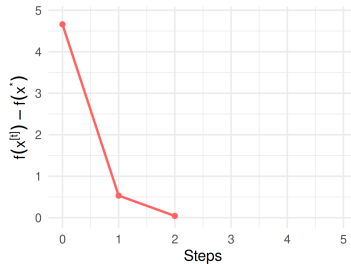
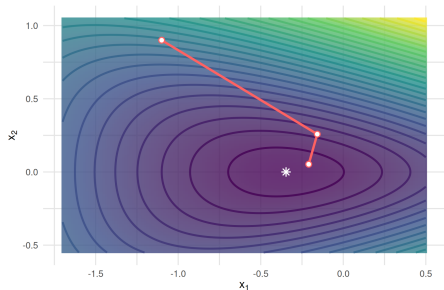


NON-QUADRATIC EXAMPLES: CONVERGENCE

Example 1: Newton-Raphson on

$$f(\mathbf{x}) = \exp(x_1 + 3x_2 - 0.1) + \exp(x_1 - 3x_2 - 0.1) + \exp(-x_1 - 0.1)$$

After 2 Newton steps:

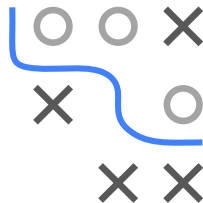
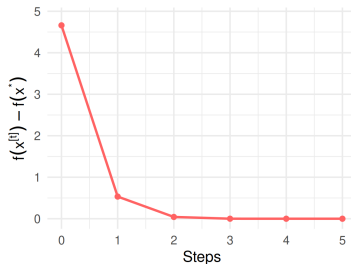
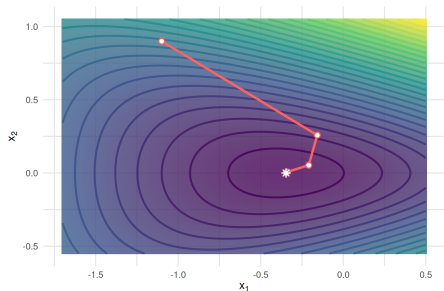


NON-QUADRATIC EXAMPLES: CONVERGENCE

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$$f(\mathbf{x}) = \exp(x_1 + 3x_2 - 0.1) + \exp(x_1 - 3x_2 - 0.1) + \exp(-x_1 - 0.1)$$

After 5 Newton steps $\Rightarrow$ Newton-Raphson converges in just 5 steps:



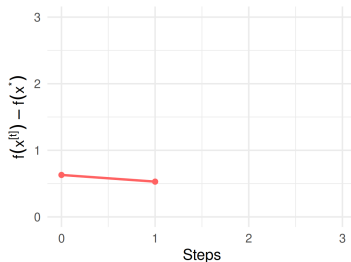
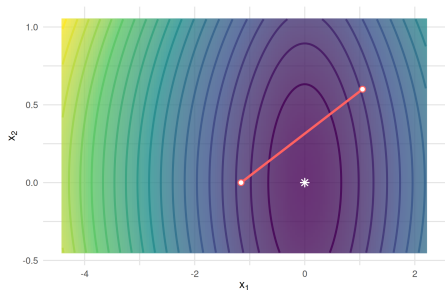
NON-QUADRATIC EXAMPLES: OVERSHOOTING

Example 2: Newton-Raphson on $f(\mathbf{x}) = (1 + x_1^2)^{1/2} + \frac{x_2^2}{2}$

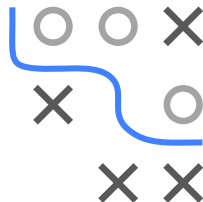
Note: along x_1 , curvature decays away from the optimum:

$$\frac{\partial^2}{\partial x_1^2} (1 + x_1^2)^{1/2} = (1 + x_1^2)^{-3/2} \rightarrow 0$$

After 1 Newton step:



$\Rightarrow f$ is quadratic in x_2 , so this coordinate is solved exactly in one step



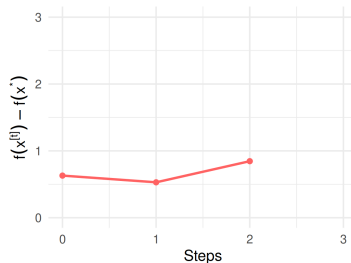
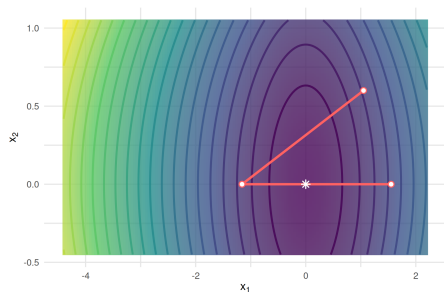
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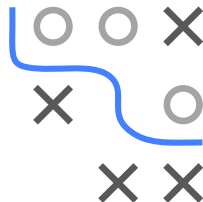
Note: along x_1 , curvature decays away from the optimum:

$$\frac{\partial^2}{\partial x_1^2} (1 + x_1^2)^{1/2} = (1 + x_1^2)^{-3/2} \rightarrow 0$$

After 2 Newton steps:



$\Rightarrow$ along x_1 , the quadratic approx. is too flat, its minimum lies well beyond $\mathbf{x}^* \Rightarrow$ NR overshoots



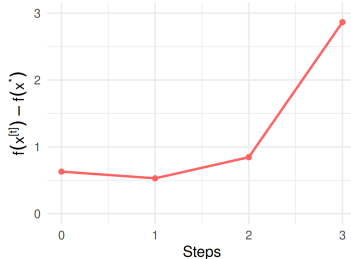
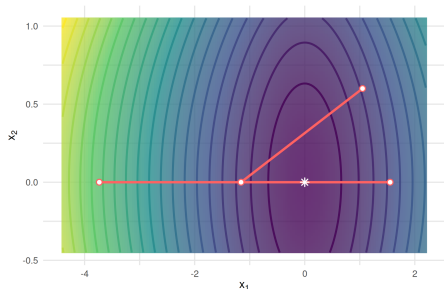
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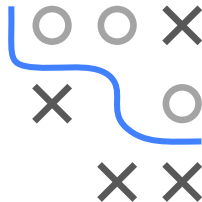
$$\frac{\partial^2}{\partial x_1^2} (1 + x_1^2)^{1/2} = (1 + x_1^2)^{-3/2} \rightarrow 0$$

After 3 Newton steps:



$\Rightarrow$ NR diverges

Note: $\mathbf{H} \succ 0$ everywhere $\Rightarrow$ direction is fine, only step size is wrong



NEWTON-RAPHSON: AFFINE INVARIANCE

- The Newton step is independent of linear (or affine) changes of coordinates
- Let $\mathbf{T} \in \mathbb{R}^{d \times d}$ be nonsingular and $\tilde{f}(\tilde{\mathbf{x}}) = f(\mathbf{T}\tilde{\mathbf{x}})$, i.e., we re-parameterize with $\mathbf{x} = \mathbf{T}\tilde{\mathbf{x}}$
- Chain rule gives

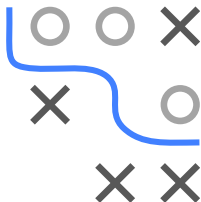
$$\nabla \tilde{f}(\tilde{\mathbf{x}}) = \mathbf{T}^T \nabla f(\mathbf{x}), \quad \nabla^2 \tilde{f}(\tilde{\mathbf{x}}) = \mathbf{T}^T \nabla^2 f(\mathbf{x}) \mathbf{T}$$

- Newton direction in the new coordinates:

$$\begin{aligned} \tilde{\mathbf{d}} &= -(\mathbf{T}^T \nabla^2 f(\mathbf{x}) \mathbf{T})^{-1} \mathbf{T}^T \nabla f(\mathbf{x}) \\ &= -\mathbf{T}^{-1} (\nabla^2 f(\mathbf{x}))^{-1} \underbrace{\mathbf{T}^{-T} \mathbf{T}^T}_{=I} \nabla f(\mathbf{x}) = \mathbf{T}^{-1} \mathbf{d} \end{aligned}$$

$\Rightarrow \mathbf{T}(\tilde{\mathbf{x}} + \tilde{\mathbf{d}}) = \mathbf{x} + \mathbf{d}$: the iterates are the same points, just expressed in different coordinates

$\Rightarrow$ NR needs no feature scaling / preconditioning (**contrast GD**: invariant only under orthogonal $\mathbf{T}$, a general $\mathbf{T}$ changes κ and thus its rate) ▶ Boyd and Vandenberghe 2004



DISCUSSION

- **Advantage:**

For f sufficiently smooth:

Newton-Raphson converges *locally* quadratically
(i.e., for starting points close enough to a stationary point)
 $\Rightarrow$ see “*Deep Dive: Convergence of Newton-Raphson*” slides

- **Disadvantage:**

For “bad” starting points:

Newton-Raphson may diverge
 $\Rightarrow$ see “*Limitations of Newton-Raphson*” slides

